

Sheet 3 — Solutions

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To be discussed on Tuesday, October 6, 16:00

Submission deadline: Monday, October 5, 23:59 (Basel time)

Please write your full name and student ID number directly on the assignment.

Problems 1–6 are required. Problem 7 is optional. An attempt at Problem 7 will be taken into account when assessing your effort, but there is no penalty for leaving it out.

Conventions. All matrices are written in the ordered computational basis unless stated otherwise. The symbol $|0\rangle$ denotes a basis vector, not the zero vector $\mathbf{0}$.

1. Computational Basis and Vector Spaces

Work in $\mathbb{C}^3$ with the orthonormal computational basis $|0\rangle = (1, 0, 0)^T$, $|1\rangle = (0, 1, 0)^T$ and $|2\rangle = (0, 0, 1)^T$.

- Write the matrices $|0\rangle\langle 0|$, $|0\rangle\langle 1|$ and $|2\rangle\langle 0|$. Give a general rule for the matrix entries of $|i\rangle\langle j|$, where $i, j \in \{0, 1, 2\}$; you do not need to write the other six matrices.
- Evaluate $\sum_{i=0}^2 |i\rangle\langle i|$.
- Which of the following sets are vector subspaces of $\mathbb{C}^3$? Justify your answers:

$$W = \{(0, z_1, z_2)^T : z_1, z_2 \in \mathbb{C}\}, \quad U = \{|v\rangle \in \mathbb{C}^3 : \langle v|v\rangle = 1\}.$$

Solution

- An outer product is a column vector multiplied by a row vector:

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad |2\rangle\langle 0| = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

More generally,

$$(|i\rangle\langle j|)_{k\ell} = \langle k|i\rangle \langle j|\ell\rangle = \delta_{ki}\delta_{j\ell}.$$

Thus the only nonzero entry is a 1 in row i , column j , with rows and columns labelled 0, 1, 2.

- The three diagonal outer products sum to the identity:

$$\sum_{i=0}^2 |i\rangle\langle i| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

This is the completeness relation.

- A vector space V obeys properties

- There is an element $\mathbf{0} \in V$, such that, for each $\mathbf{v} \in V$, $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$ (existence of neutral element)

- For each $\mathbf{v} \in V$, there exists $\mathbf{u} = (-1)\mathbf{v}$ in V such that $\mathbf{v} + \mathbf{u} = \mathbf{0}$ (existence of inverse element)

W contains $\mathbf{0}$ and is closed under addition and multiplication by any complex scalar: its first coordinate remains zero. Moreover, $W = \text{span}\{|1\rangle, |2\rangle\}$, so $\dim_{\mathbb{C}} W = 2$. The set U is not a subspace: it does not contain $\mathbf{0}$ and is not closed under scalar multiplication. For example, $|0\rangle \in U$ but $2|0\rangle \notin U$ because its norm is 2. Normalised vectors belong to a Hilbert space, but they do not themselves form a vector space.

2. Inner Products and Norms

Let $|v\rangle = (1, i)^T$ and $|w\rangle = (2, 1 - i)^T$ in $\mathbb{C}^2$.

- Write $\langle v|$ and $\langle w|$, compute $\langle v|w\rangle$ and $\langle w|v\rangle$, and verify the conjugate-symmetry relation $\langle w|v\rangle = \langle v|w\rangle^*$.
- Find the norms of both vectors and write the normalised vectors $|\hat{v}\rangle$ and $|\hat{w}\rangle$ explicitly.

Solution

- Taking the conjugate transpose gives $\langle v| = (1, -i)$ and $\langle w| = (2, 1 + i)$. Hence

$$\begin{aligned}\langle v|w\rangle &= 2 - i(1 - i) = 1 - i, \\ \langle w|v\rangle &= 2 + (1 + i)i = 1 + i = \langle v|w\rangle^*.\end{aligned}$$

- Since $\langle v|v\rangle = 1 + |i|^2 = 2$ and $\langle w|w\rangle = 4 + |1 - i|^2 = 6$,

$$\|v\| = \sqrt{2}, \quad \|w\| = \sqrt{6}, \quad |\hat{v}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |\hat{w}\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 - i \end{pmatrix}.$$

Dividing by the norm makes each inner product with itself equal to 1.

3. Bra-Ket Algebra

Let $|u\rangle, |v\rangle, |w\rangle \in \mathbb{C}^n$ be unit vectors and let $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear operator. The vectors are not assumed to be mutually orthogonal. Simplify each expression and state whether its result is a scalar, a ket vector or an operator:

- $(|u\rangle\langle v|)(|v\rangle\langle w|)$;
- $(\langle u|A|v\rangle)|w\rangle$;
- $(|u\rangle\langle u|)^2$;
- $\langle w|(|u\rangle\langle v|)|w\rangle$.

Solution

- (a) Contract the adjacent bra and ket:

$$(|u\rangle\langle v|)(|v\rangle\langle w|) = |u\rangle\langle v|v\rangle\langle w| = |u\rangle\langle w|.$$

This is an operator. Only the normalisation of $|v\rangle$ was needed.

- (b) $\langle u|A|v\rangle$ is a scalar, so the result is the ket vector $\langle u|A|v\rangle |w\rangle$. It may be the zero vector if the scalar vanishes.

- (c) Since $\langle u|u\rangle = 1$,

$$(|u\rangle\langle u|)^2 = |u\rangle\langle u|u\rangle\langle u| = |u\rangle\langle u|.$$

The result is an operator.

- (d) The result is the scalar $\langle w|u\rangle\langle v|w\rangle$. Unlike $\langle w|w\rangle$, this product is not generally an absolute square and need not be real.

To check the type, treat a ket as an $n \times 1$ matrix, a bra as a $1 \times n$ matrix and an operator as an $n \times n$ matrix. In particular, $\langle v|w\rangle$ is a scalar whereas $|v\rangle\langle w|$ is an operator: counting bras and kets alone does not determine the result. Scalars can be moved past other factors, but the order of matrix factors must be preserved.

4. Qubits, Measurement and Antipodal States

A pure qubit state, up to a global phase, can be written as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi < 2\pi.$$

Its Bloch vector is $\mathbf{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. Recall the Born rule: in an orthonormal measurement basis $\{|a_j\rangle\}$, the probability of outcome j is $|\langle a_j|\psi\rangle|^2$.

- (a) Starting from two complex amplitudes $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, explain why a pure qubit state has two independent real parameters. Compute the computational-basis measurement probabilities. Show that multiplying $|\psi\rangle$ by $e^{i\gamma}$, with $\gamma \in \mathbb{R}$, changes no measurement probability in any orthonormal basis.
- (b) Using Dirac notation, show that antipodal Bloch vectors $\mathbf{r}$ and $-\mathbf{r}$ correspond to orthogonal states.
- (c) Let $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Compare their measurement probabilities in the computational basis and in the basis $\{|+\rangle, |-\rangle\}$.

Solution

- (a) The two complex amplitudes contain four real parameters. The constraint $|\alpha|^2 + |\beta|^2 = 1$ removes one, and identifying vectors that differ only by a global phase removes another. Thus two real parameters remain. The angles above describe them, with the usual coordinate degeneracy at the poles.

Applying the Born rule gives

$$p(0) = |\alpha|^2 = \cos^2\frac{\theta}{2}, \quad p(1) = |\beta|^2 = \sin^2\frac{\theta}{2}, \quad p(0) + p(1) = 1.$$

After outcome 0 or 1, the state is respectively $|0\rangle$ or $|1\rangle$, up to a global phase. For any unit basis vector $|a_j\rangle$,

$$|\langle a_j | (e^{i\gamma} |\psi\rangle)|^2 = |e^{i\gamma}|^2 |\langle a_j | \psi\rangle|^2 = |\langle a_j | \psi\rangle|^2.$$

This is why a global phase has no observable effect on these probabilities.

- (b) The antipodal point has angles $(\pi - \theta, \phi + \pi)$, with the azimuth understood modulo 2π . A normalised representative is

$$\begin{aligned} |\psi_\perp\rangle &= \cos \frac{\pi - \theta}{2} |0\rangle + e^{i(\phi + \pi)} \sin \frac{\pi - \theta}{2} |1\rangle \\ &= \sin \frac{\theta}{2} |0\rangle - e^{i\phi} \cos \frac{\theta}{2} |1\rangle. \end{aligned}$$

Writing $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$, we obtain

$$\langle \psi | \psi_\perp \rangle = cs - e^{-i\phi} e^{i\phi} sc = 0.$$

Thus antipodal Bloch points represent orthogonal states, not state vectors that differ merely by an overall minus sign.

- (c) For either $|+\rangle$ or $|-\rangle$, the computational-basis probabilities are $p(0) = p(1) = 1/2$. In the $\{|+\rangle, |-\rangle\}$ basis,

input	$p(+)$	$p(-)$
$ +\rangle$	1	0
$ -\rangle$	0	1

because $\langle + | - \rangle = 0$ and both states have unit norm.

5. Bloch-Sphere Coordinates

Use the parametrisation and angle ranges in Problem 4. The Pauli matrices are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (a) Find (θ, ϕ) and the Bloch vector for each of the states

$$|0\rangle, \quad |1\rangle, \quad |+\rangle, \quad |-\rangle, \quad \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

- (b) Find a normalised ket, in the standard Bloch form, corresponding to $\mathbf{r} = (1, 1, 1)/\sqrt{3}$. Give its amplitudes exactly, not only as decimal approximations.

- (c) By explicit matrix multiplication, show that for a normalised pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$(\langle \psi | X | \psi \rangle, \langle \psi | Y | \psi \rangle, \langle \psi | Z | \psi \rangle) = (2 \operatorname{Re}(\alpha^* \beta), 2 \operatorname{Im}(\alpha^* \beta), |\alpha|^2 - |\beta|^2).$$

Substitute the Bloch parametrisation and show that this is precisely $\mathbf{r}$.

Solution

- (a) Comparing the amplitudes with the standard form gives the following table. “Arbitrary” means that all choices describe the same physical state.

State	θ	ϕ	Bloch vector
$ 0\rangle$	0	arbitrary	$(0, 0, 1)$
$ 1\rangle$	π	arbitrary	$(0, 0, -1)$
$ +\rangle$	$\pi/2$	0	$(1, 0, 0)$
$ -\rangle$	$\pi/2$	π	$(-1, 0, 0)$
$(0\rangle + i 1\rangle)/\sqrt{2}$	$\pi/2$	$\pi/2$	$(0, 1, 0)$
$(0\rangle - i 1\rangle)/\sqrt{2}$	$\pi/2$	$3\pi/2$	$(0, -1, 0)$

At the north pole, $\sin(\theta/2) = 0$, so the amplitude containing ϕ vanishes. At the south pole, $\cos(\theta/2) = 0$ and the ket is $e^{i\phi}|1\rangle$: here ϕ is an overall phase. The two explanations are different, but in both cases ϕ is physically irrelevant. To obtain exactly the ket $|1\rangle$, one may choose $\phi = 0$.

- (b) From $r_z = \cos \theta = 1/\sqrt{3}$,

$$\theta = \arccos \frac{1}{\sqrt{3}}, \quad \sin \theta = \sqrt{\frac{2}{3}}.$$

Since both r_x and r_y are positive and equal, $\cos \phi = \sin \phi = 1/\sqrt{2}$, so $\phi = \pi/4$. The half-angle identities then give

$$|\psi\rangle = \sqrt{\frac{1+1/\sqrt{3}}{2}} |0\rangle + e^{i\pi/4} \sqrt{\frac{1-1/\sqrt{3}}{2}} |1\rangle.$$

The squared moduli sum to 1, and this representative has a positive real $|0\rangle$ amplitude. Any overall phase describes the same state.

- (c) Multiplying the matrices by $(\alpha, \beta)^T$ gives

$$\begin{aligned} \langle\psi|X|\psi\rangle &= (\alpha^*, \beta^*) \begin{pmatrix} \beta \\ \alpha \end{pmatrix} = \alpha^* \beta + \beta^* \alpha = 2 \operatorname{Re}(\alpha^* \beta), \\ \langle\psi|Y|\psi\rangle &= (\alpha^*, \beta^*) \begin{pmatrix} -i\beta \\ i\alpha \end{pmatrix} = -i\alpha^* \beta + i\beta^* \alpha = 2 \operatorname{Im}(\alpha^* \beta), \\ \langle\psi|Z|\psi\rangle &= (\alpha^*, \beta^*) \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = |\alpha|^2 - |\beta|^2. \end{aligned}$$

Now set $\alpha = c = \cos(\theta/2)$ and $\beta = e^{i\phi}s$ with $s = \sin(\theta/2)$. Then

$$\mathbf{r} = (2cs \cos \phi, 2cs \sin \phi, c^2 - s^2) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

Thus the three coordinates are the expectation values of the Pauli observables. In particular, they are real and $\|\mathbf{r}\| = 1$ for a pure qubit state.

6. Linear Operators and Their Matrix Representations

Let $B : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear operator satisfying $B|j\rangle = |\psi_j\rangle$ for $j = 0, \dots, n-1$. The vectors $|\psi_j\rangle$ need not be normalised or linearly independent.

- (a) Express B as a sum of outer products. Explain why its action on a basis determines its action on every vector.
- (b) Show that the coordinate column of $|\psi_j\rangle$ is column j of the matrix of B .
- (c) For the Hadamard operator

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

find $H|0\rangle$, $H|1\rangle$ and an outer-product expression for H . Use linearity to find $H(\alpha|0\rangle + \beta|1\rangle)$.

Solution

- (a) Insert the completeness relation:

$$B = BI_n = B \sum_{j=0}^{n-1} |j\rangle\langle j| = \sum_{j=0}^{n-1} |\psi_j\rangle\langle j|.$$

Every vector has a unique expansion $|v\rangle = \sum_j v_j |j\rangle$. Linearity therefore fixes its image as $B|v\rangle = \sum_j v_j |\psi_j\rangle$, so no further information about B is needed. This also verifies the outer-product expression directly.

- (b) The matrix elements are

$$B_{ij} = \langle i|B|j\rangle = \langle i|\psi_j\rangle.$$

For fixed j , these are exactly the components of $|\psi_j\rangle$ as i runs down column j . Equivalently, $B = [\psi_0 \ \psi_1 \ \dots \ \psi_{n-1}]$, with each ψ_j written as a coordinate column.

- (c) The two columns give

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

Thus

$$\begin{aligned} H &= |+\rangle\langle 0| + |-\rangle\langle 1| \\ &= \frac{1}{\sqrt{2}}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|), \\ H(\alpha|0\rangle + \beta|1\rangle) &= \alpha|+\rangle + \beta|-\rangle = \frac{\alpha + \beta}{\sqrt{2}}|0\rangle + \frac{\alpha - \beta}{\sqrt{2}}|1\rangle. \end{aligned}$$

7. Mixed States and the Bloch Ball (Optional)

Background for this optional problem. If a preparation produces normalised states $|\psi_k\rangle$ with classical probabilities $q_k \geq 0$, $\sum_k q_k = 1$, its density matrix is

$$\rho = \sum_k q_k |\psi_k\rangle\langle\psi_k|.$$

The probability of an outcome associated with a normalised vector $|a\rangle$ in an orthonormal measurement basis is

$$p(a) = \langle a|\rho|a\rangle = \sum_k q_k |\langle a|\psi_k\rangle|^2.$$

A qubit density matrix has the Bloch representation

$$\rho = \frac{1}{2}(I_2 + r_x X + r_y Y + r_z Z).$$

Here $\mathbf{r} \in \mathbb{R}^3$, and physical states fill the Bloch ball $\|\mathbf{r}\| \leq 1$; pure states lie on its surface.

- A preparation produces $|+\rangle$ with probability p and $|-\rangle$ with probability $1-p$, where $0 \leq p \leq 1$. Write its density matrix explicitly, find its Bloch vector and locate it in the Bloch ball. For which values of p is the state pure?
- At $p = 1/2$, compare this mixture with the coherent superposition $|\chi\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ by computing the computational-basis measurement probabilities for both preparations.
- Why can a vector $\mathbf{r}$ with $\|\mathbf{r}\| > 1$ not represent a physical qubit state? Consider the matrix $\rho_{\mathbf{r}} = (I_2 + r_x X + r_y Y + r_z Z)/2$. *Hint:* Choose a pure state $|a\rangle$ with Bloch vector $-\mathbf{r}/\|\mathbf{r}\|$, and use Problem 5(c) to compute $\langle a|\rho_{\mathbf{r}}|a\rangle$.

Solution

- Since

$$|+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |-\rangle\langle-| = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

their probabilistic mixture is

$$\rho = p|+\rangle\langle+| + (1-p)|-\rangle\langle-| = \frac{1}{2} \begin{pmatrix} 1 & 2p-1 \\ 2p-1 & 1 \end{pmatrix} = \frac{1}{2}(I_2 + (2p-1)X).$$

Hence $\mathbf{r} = (2p-1, 0, 0)$: the point lies on the line segment between the two x -axis poles. At $p = 0$ and $p = 1$, it is respectively the pure state $|-\rangle$ or $|+\rangle$. For $0 < p < 1$, $\|\mathbf{r}\| = |2p-1| < 1$, so it is mixed and cannot be represented by a single normalised ket. At $p = 1/2$, $\rho = I_2/2$ and $\mathbf{r} = \mathbf{0}$.

- For the mixture at $p = 1/2$,

$$p_{\text{mix}}(0) = \langle 0|I_2/2|0\rangle = \frac{1}{2}, \quad p_{\text{mix}}(1) = \frac{1}{2}.$$

In contrast,

$$|\chi\rangle = \frac{1}{\sqrt{2}} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |0\rangle,$$

so $p_\chi(0) = 1$ and $p_\chi(1) = 0$. The superposition has interference terms:

$$|\chi\rangle\langle\chi| = \frac{1}{2}(|+\rangle\langle+| + |-\rangle\langle-| + |+\rangle\langle-| + |-\rangle\langle+|),$$

while the mixture has only the first two terms. A classical mixture averages probabilities, not amplitudes. In fact, $I_2/2$ gives probability $1/2$ for each outcome in every orthonormal qubit basis, since $\langle a|I_2/2|a\rangle = 1/2$ for every unit $|a\rangle$.

- (c) Write $R = \|\mathbf{r}\| > 1$ and choose a normalised pure state whose Bloch vector is $\mathbf{n} = -\mathbf{r}/R$. Such a state exists because $\|\mathbf{n}\| = 1$. By Problem 5(c), its Pauli expectation values are n_x, n_y, n_z . Therefore,

$$\begin{aligned}\langle a|\rho_{\mathbf{r}}|a\rangle &= \frac{1}{2}(1 + r_x n_x + r_y n_y + r_z n_z) \\ &= \frac{1}{2}\left(1 - \frac{r_x^2 + r_y^2 + r_z^2}{R}\right) = \frac{1 - R}{2} < 0.\end{aligned}$$

This would be a negative probability for the outcome $|a\rangle$ in a basis containing it, which is impossible. Thus points outside the Bloch ball cannot represent physical qubit states in this representation.